

NISHANTH ANANTHA

Principal Technical Program Manager · AI Systems Builder · Portfolio Governance · Edge AI
nishanth.anantha87@gmail.com · +1 (213) 327-7697 · San Marcos, CA · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

ABOUT

Staff TPM with 14+ years owning cross-functional portfolio visibility and delivery accountability across hardware, embedded SW, and AI systems programs. Builder, Connector, Translator: I build production AI systems (Claude API + MCP + ensemble ML), connect HW/FW/SW/AI teams around a shared execution reality, and translate business commitments into engineering capacity plans — surfacing tradeoffs before they become misses. Technical depth spans FPGA/embedded systems, SoC integration, and production agentic AI — enough to engage credibly with hardware, embedded, cloud, and computer vision teams without domain specialists present.

WHAT I'VE DELIVERED

- Owned portfolio-level program governance across 12+ concurrent NPI programs — single source of truth for commitments, capacity, dependencies, and status; surfaced cross-team blockers to Director and VP before they became schedule misses; 29%→94% release success rate
- Built production agentic AI portfolio intelligence system (Claude API + MCP + GitHub + JIRA + Smartsheet + ensemble ML) — autonomous signal collection across hardware and SW programs with SHAP explainability; adopted by Director and VP stakeholders; 40% release risk reduction
- Translated business commitments into engineering capacity plans — chaired cross-ocean CCB coordinating HW/FW/SW programs across US, Europe, and Asia; managed rapid engineering change velocity and prioritization decisions when capacity and demand conflicted
- Managed \$2M+ validation infrastructure from scratch — vendor qualification, procurement, and platform delivery with 70% field defect reduction; \$3M+ customer savings via rapid cross-functional escalation resolution
- Built Edge CV Inference System (YOLOv3 + FastAPI + Docker) — person/vehicle detector with p50/p95 latency telemetry, cold-start benchmarking (13.9x penalty), and confidence threshold analysis; mirrors production edge AI deployment stack

EXPERIENCE

Staff Technical Program Manager — Portfolio Intelligence & NPI Governance Jan 2024 – Present

ASML · San Diego, CA · Team: 20

- Owned company-level program portfolio — maintained single source of truth for commitments, capacity, dependencies, and status across all concurrent NPI programs; designed executive reporting cadences giving VP leadership clear visibility without noise
- Built and deployed production agentic AI portfolio intelligence system (Claude API + MCP + GitHub REST + JIRA + Smartsheet + PyTorch + GBM + DistilBERT + SHAP + FastAPI) — security hardened; feeds Go/NoGo operating reviews adopted by Director and VP
- Served as primary interface between executive leadership and engineering — translated business commitments into engineering capacity plans; surfaced tradeoffs before they became misses; drove prioritization when capacity and demand conflicted

Sr. Technical Program Manager — SW/HW Integration & Platform Programs Mar 2019 – Dec 2023

ASML · San Diego, CA · Team: 10

- Built AI program operating model from scratch — execution forums, Change Control Board, release cadences, dependency tracking, and portfolio health reporting; introduced governance in a high-growth engineering environment
- Chaired cross-ocean CCB — coordinated HW/FW/SW change velocity across US, Europe, and Asia; translated engineering change impact into business tradeoffs for senior leadership weekly
- Drove CI/CD transformation org-wide (Jenkins, GitHub Actions, Docker) — 65% faster delivery; built ensemble ML risk predictor (PyTorch + GBM + DistilBERT, 13 signals, SHAP); 35% reduction in last-minute integration failures
- Reduced customer-facing defects 50%; resolved \$3M+ in customer escalations; managed \$2M+ validation infrastructure with 70% field defect reduction

Sr. Software Engineer / Team Lead / Scrum Master Nov 2016 – Mar 2019

ASML · San Diego, CA · Team: 7

- Engineer-to-TPM arc: shipped Python automation to production while owning SAFe Agile planning, backlog management, and release governance
- Lifted release success from 29% to 94% in 18 months; resolved \$3M+ customer escalation; awarded for Leadership and Ownership

SWI Consultant — NPI Bring-Up Jan 2021 – Jun 2021

HMI (acq. by ASML) · San Jose, CA

- Built entire execution framework from zero in high-growth startup — portfolio baseline, dependency map, Agile cadence, validation coverage 50%→80%; delivered major customer commitment one month early

Sr. Electrical Engineer / SW Engineer — Embedded & Edge Systems 2009 – 2016

Caterpillar · Volvo Group · Abbott

- Caterpillar/ARI: Embedded C on dSPACE ATEs (Freescale MPC535/SoC); PWM/complex I/O HIL testbenches; Lauterbach firmware debugging; Python automation; \$500K+ field escalation resolution.
- Volvo Group: FPGA-based cRIO LabVIEW RT ATE simulating vehicle powertrain environments -- CAN J1939/J1587, NI TestStand; global owner of Bluetooth infotainment SW test platform. Patent Idea of the Year 2012 - Volvo Global Patent Council.
- Abbott: LabVIEW CVI ATE testbench for heart pacemaker IC/wafer characterization; GPIB/VISA instrument drivers; FDA-compliant documentation.

AI SYSTEMS & FEATURED BUILDS

Agentic Portfolio Intelligence System

Claude API · MCP · GitHub REST · JIRA · Smartsheet · PyTorch · GBM · DistilBERT · SHAP · FastAPI

- Agentic Claude loop routes tool calls across live operational data sources — synthesizing 13 execution signals into portfolio risk score with SHAP factor attribution; security hardened; feeds executive operating reviews; adopted in production

Multi-Agent System — NVIDIA NeMo Agent Toolkit (NAT) + LangGraph

NVIDIA NeMo Agent Toolkit · LangGraph · Multi-Agent Orchestration · Agent Evaluation

- Built and deployed a multi-agent workflow on NVIDIA's NeMo Agent Toolkit with LangGraph orchestration — designed agent handoff logic and an evaluation harness to benchmark task completion and routing accuracy
- Extends production agentic system experience (Claude API + MCP) into NVIDIA-native tooling — direct, hands-on grounding for infrastructure- and platform-specific technical conversations

Edge CV Inference System — AI Builder Project

YOLOv3 · cvlib · FastAPI · Docker · TensorFlow 2.15 · Python 3.11

- YOLOv3 person/vehicle detector deployed as versioned FastAPI service with structured telemetry (device shadow analog), p50/p95 latency tracking per inference, and person/vehicle/street object filtering
- Benchmarked cold-start penalty (13.9x: 1073ms → 77ms cached), confidence threshold operating point (0.3 validated), and multi-stream architecture constraints; documented production path to NVIDIA DeepStream / Jetson NPU

RAG Conversational AI Agent with Voice Evaluation Pipeline

HuggingFace · ChromaDB · W&B LLM Evaluation · Deepgram STT/TTS · Streaming Inference

- End-to-end RAG system with voice pipeline; offline eval (BLEU/ROUGE, W&B LLM Evaluation) → online (82% CSAT, 65% deflection, 1.1s p50); multi-modal evaluation with Deepgram STT/TTS

SKILLS & TOOLS

- **Portfolio Governance:** Cross-functional portfolio management, capacity planning, demand management, executive reporting, operating rhythm design, single source of truth systems, dependency management, PMI-PMP, Agile/SAFe
- **Technical Fluency:** Hardware NPI, embedded SW/HW (FPGA, cRIO, LabVIEW RT, CAN), SoC platforms, AI/ML systems, CI/CD, edge-deployed systems, real-time systems — credible across all domains without specialists present
- **AI / ML Stack:** Claude API (Anthropic certified), Agentic AI, MCP, LangGraph, NVIDIA NeMo Agent Toolkit, PyTorch, HuggingFace, RAG, ChromaDB, SHAP, Google Vertex AI
- **Tools & Platforms:** JIRA, Smartsheet, Confluence, MS Project, Python, GitHub Actions, Jenkins, Docker, FastAPI, REST, AWS Certified, SQL, Power BI

EDUCATION

M.S. Electrical & Electronics Engineering — University of Southern California (USC), 2014

B.E. Electronics & Communication — SJCE / VTU, India, 2009

CERTIFICATIONS

Building with Claude API — Anthropic (Apr 2026) · Claude on Google Cloud Vertex AI — Anthropic (May 2026) · NVIDIA NeMo Agent Toolkit — NVIDIA (Jul 2026) · Red Teaming LLM Apps — Giskard (Jun 2026) · Safe and Reliable AI via Guardrails — Guardrails.AI (Jun 2026) · Machine Learning in Production (MLOps) — DeepLearning.AI (Jun 2026) · PMI-PMP (Active) · AWS Certified Cloud Practitioner (2024)